

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicentre, Randomised, Double-blind, Chronic-dosing, Parallel-group, Placebo-controlled Extension Study to Evaluate the Long-term Efficacy and Safety of Tozorakimab in Participants with Chronic Obstructive Pulmonary Disease (COPD) with a History of Exacerbations **Short Title/ Acronym:** PROSPERO **Protocol Number:** D9180C00008 **NCT Number:** NCT05742802 **Sponsor Name:** AstraZeneca **Investigational Product:** Tozorakimab (Anti-IL-33 monoclonal antibody) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and efficacy of tozorakimab in reducing the annualized rate of moderate-to-severe COPD exacerbations over a combined 104-week period in former smokers. **Secondary Objectives:** To assess the annualized rate of moderate-to-severe exacerbations in the overall population (former and current smokers), evaluate long-term changes in lung function (e.g., pre-bronchodilator FEV1), and continuously monitor long-term safety and tolerability.

2.2 Background & Rationale To address the significant unmet need in COPD where up to half of patients remain at high risk of exacerbations, hospitalizations, cardiopulmonary events, and death despite standard maintenance therapy. IL-33 is released by airway cells during cellular stress, driving inflammation and mucus dysfunction. Tozorakimab neutralizes both reduced and oxidized forms of IL-33. The PROSPERO trial functions as a long-term extension for patients who successfully completed the predecessor trials (OBERON or TITANIA) to capture 2-year efficacy and safety data.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Extension Study) **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Parallel-group (1:1 re-randomization for previous placebo participants; active arm participants maintain their assigned dose)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 1,713 participants **Number of Sites:** Multicentre (Global) **Estimated**

Study Start: Q3 2023 **Estimated Primary Completion:** First half of 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants who have completed the 52-week treatment period and have not been prematurely discontinued from the investigational product (IP) in the predecessor studies (OBERON or TITANIA). 2. Received their last dose of IP in the predecessor studies within the previous 12 weeks and were not withdrawn. 3. Female participants of childbearing potential (FOCBP) must have a negative urine pregnancy test at Visit 1 and be willing to continue using contraceptive methods as agreed to for the predecessor studies.

4.2 Exclusion Criteria 1. Any clinically significant disorder, abnormal findings, or major physical/cognitive impairment that may put the participant at risk or impact the interpretation of the study results. 2. Concurrent enrolment in other interventional clinical studies or treatment with another IP (with the exception of the predecessor studies). 3. Known history of severe allergic reaction to any monoclonal antibody or IP component, or chronic use of immunosuppressive medications (e.g., systemic corticosteroids).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tozorakimab dose 1 or dose 2, administered once every 4 weeks. Participants re-randomized to placebo will receive a matching volume. **Route of Administration:** Subcutaneous (SC) injection via a pre-filled syringe. **Duration of Treatment:** 52 weeks (totaling up to 104 weeks of continuous therapy when combined with predecessor study exposure).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care (SoC) inhaled COPD maintenance therapy (dual or triple therapy). **Prohibited:** Chronic use of systemic immunosuppressive medications, marketed or investigational biologics outside of the study protocol.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening / Visit 1** | Day -14 to 0 | Transition consent from OBERON/TITANIA, Pregnancy Test, Eligibility Confirmation | | **Baseline** | Day 0 | Re-randomization (for placebo roll-overs) or Dose Continuation, SC Injection | | **Interim** |
Every 4 Weeks | SC Injections, Safety Labs, Lung Function (FEV1) Testing, Exacerbation Review | | **End of Study** | Week 52 | Final

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized severe exacerbation rate over 104 weeks in former smokers (combining data from OBERON/TITANIA and the PROSPERO extension phase). **Measurement Method:** Clinical assessment and tracking of moderate-to-severe exacerbations requiring systemic corticosteroids, antibiotics, and/or hospital admission.

7.2 Secondary & Exploratory Endpoints * Annualized rate of moderate-to-severe exacerbations in the overall population (current and former smokers). * Change from baseline in pre-bronchodilator forced expiratory volume in 1 second (FEV1). * Safety and tolerability profile (incidence of treatment-emergent adverse events over 104 weeks).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via site visits, physical checks, and patient electronic diaries (e-Diaries). **Serious Adverse Events (SAEs):** Standard reporting timelines to AstraZeneca and the ethics committees (e.g., within 24 hours of investigator awareness). **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee reviewing unblinded safety data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy endpoints. Safety Set for AE/SAE analysis. **Sample Size Justification:** The sample size (\$N \approx 1,713\$) is determined by the number of eligible rollover participants from the Phase 3 OBERON and TITANIA trials to adequately evaluate long-term exposure. **Handling of Missing Data:** Advanced imputation models (e.g., Multiple Imputation or Mixed Model for Repeated Measures) for continuous efficacy outcomes such as lung function over the 52-week extension.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to assure adherence to the protocol and data integrity. **Audit Trail:**

Continuous eCRF monitoring, data clarification forms, and restricted unblinding protocols.

11. Study Identifiers & Governance

Primary ID: D9180C00008 **Registry Number:** NCT05742802 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PROSPERO **Official Regulatory Title:** A Phase III, Multicentre, Randomised, Double-blind, Chronic-dosing, Parallel-group, Placebo-controlled Extension Study to Evaluate the Long-term Efficacy and Safety of Tozorakimab in Participants with Chronic Obstructive Pulmonary Disease (COPD) with a History of Exacerbations

12. Clinical Framework (COPD Specific)

Primary Condition: Chronic Obstructive Pulmonary Disease (COPD) **Diagnosis Threshold:** History of symptomatic COPD and frequent exacerbations despite maintenance therapy **Medical Methodology:** Add-on biologic therapy (IL-33 inhibitor) to background standard-of-care (SoC) inhaled therapies **Optimization Target:** Reduction of annualized moderate-to-severe exacerbation rates and prevention of lung function decline

13. Study Procedures & Methodology

Management Pathway: Subcutaneous biologic administration every 4 weeks combined with standard maintenance inhaler adherence **Education Protocol (HCP):** Site training on IP handling, blinding protection, and pre-filled syringe administration **Education Protocol (Patient):** Protocol compliance, e-Diary tracking for symptoms (Breathlessness, Cough, and Sputum Scale), and tracking rescue medication usage **Quality Audit Mechanism:** Source data verification and continuous medical monitoring of exacerbation events

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (Extension) **Onsite Visit Cadence:** Every 4 weeks (Q4W) for injections and assessment **Remote Monitoring Frequency:** Between dosing visits or as needed based on symptom reporting **Checklist Review Interval:** Routine e-Diary symptom checks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |